

Roam.io

Topic: Tool Augmentation

Meet the crew:

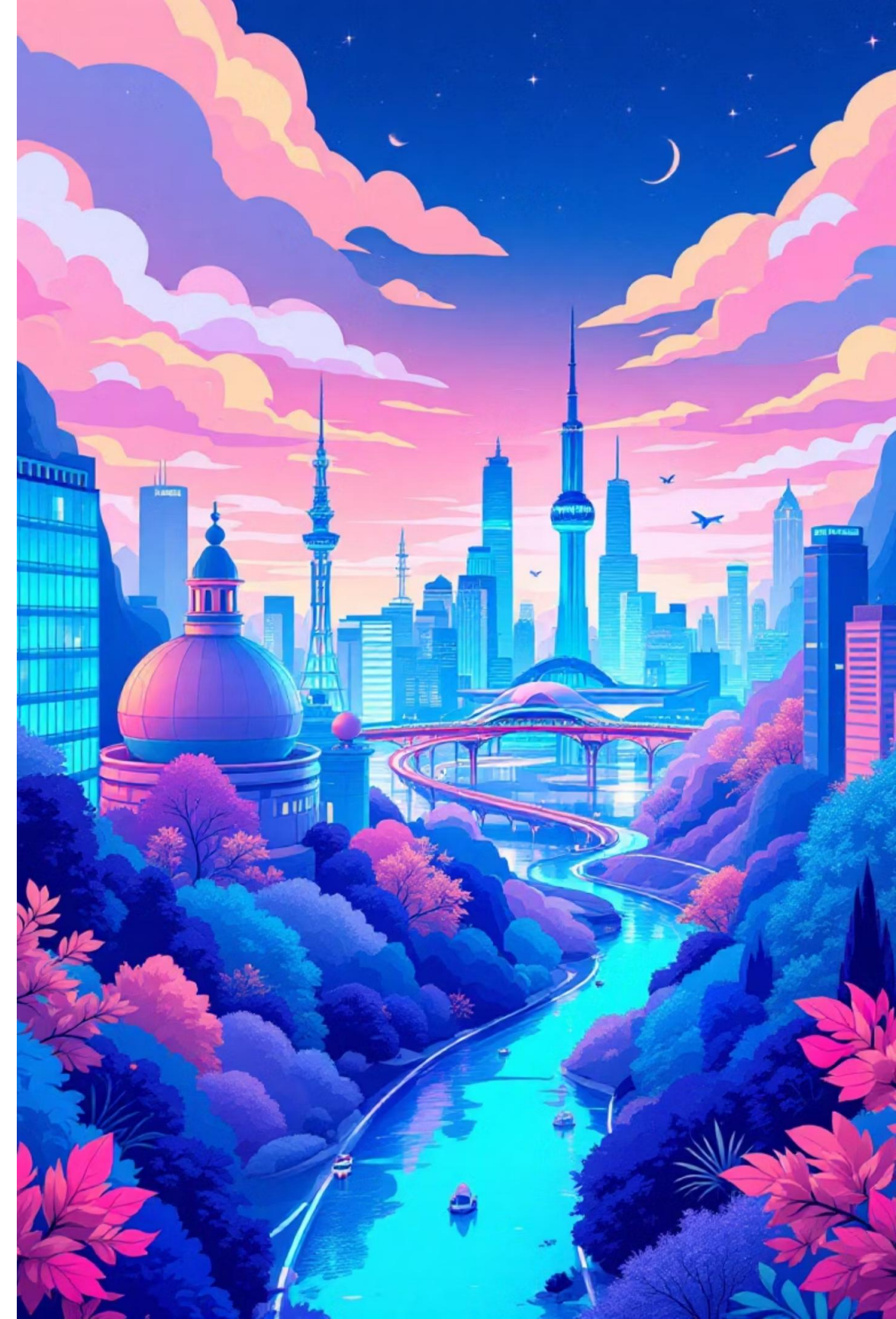
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A Problem Statement

Endless Research

Hours spent sifting through flights, hotels, activities, and dining options can be overwhelming.

Lack of Personalization

Existing platforms offer generic suggestions, failing to cater to individual budgets, unique preferences, or travel durations.

Manual Effort

The current process demands significant manual effort to piece together a cohesive and enjoyable itinerary.

Vision

To build an AI-powered multi-agent travel assistant that curates personalized surprise trips and automatically notifies users when it's the perfect time to travel based on weather, budget, and preferences.



Our Objective: An Intelligent Travel Companion



Dynamic Itineraries

Generates flexible, day-by-day plans tailored to your journey.



Preference-Based Filtering

Filters activities to perfectly match your interests, ensuring every experience is desirable.



Hidden Gems & Dining

Suggests unique restaurants and local secrets for an authentic experience.



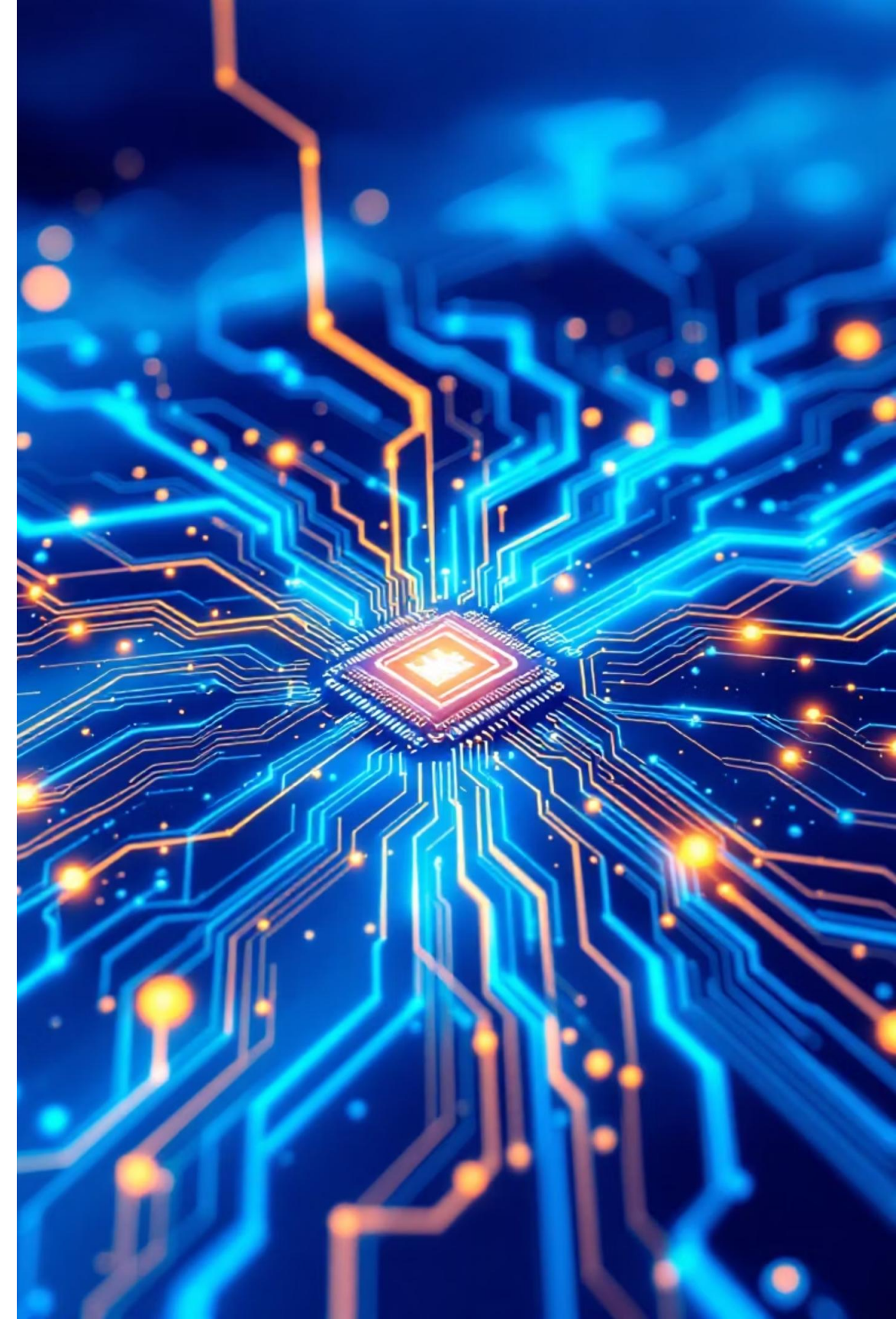
Contextual Planning

Considers real-time weather, optimal timings, and travel feasibility for seamless exploration.

Roam.io System Architecture



Our robust architecture ensures a seamless flow from user input to a fully generated, personalized itinerary.



Core Technologies Powering Roam.io

Programming & Development

- Python 3.11: The backbone for all our scripting and logic.
- Poetry: Ensures robust dependency management and project stability.

AI & LLM Integration

- GPT-4o (OpenAI): The advanced large language model driving intelligent decisions.
- CrewAI: For orchestrating our multi-agent system efficiently.
- CrewAI-Tools: Specialized tools like SerperDevTool and ScrapeWebsiteTool for real-time data.

These technologies combine to create a powerful and flexible travel planning solution.

Intelligent Agent Design with CrewAI



Personalized Activity Planner Agent

Powered by GPT-4o, this agent recommends attractions, activities, and optimal timings, using search and scraping for real-time data.



Restaurant Scout Agent

Leverages GPT-4o to filter restaurant options, analyzing cuisines, ratings, and ambiance, providing detailed reasoning for each suggestion.



Itinerary Compiler Agent

Compiles all gathered information into a final, formatted itinerary. It summarizes the day-wise plan using GPT-4o and outputs both JSON and a text summary.

Our three autonomous agents collaborate seamlessly to craft your perfect trip.

Seamless Workflow Execution



User Input

The user provides essential details: desired city, travel dates, and cuisine preferences.



Data Acquisition & Analysis

Our agents fetch and meticulously analyze real-time data, including weather and local attractions.



High-Level Reasoning

GPT-4o performs complex reasoning to interconnect all aspects of the trip for optimal planning.



Sequential Task Execution

Tasks are executed sequentially and collaboratively using a CrewAI workflow, ensuring efficiency and accuracy.



Final Itinerary Production

A comprehensive itinerary is produced in user-friendly formatted text.

This streamlined process guarantees a personalized and efficient travel plan.

Robust Tools & Frameworks

OpenAI GPT-4o

Exceptional reasoning, speed, and low latency make it ideal for intricate multi-agent travel planning scenarios.

CrewAI & CrewAI-Tools

Orchestrates agents and tasks with precision, providing essential tools for search, scraping, and data extraction.

Python & Poetry

The reliable foundation for our development environment, ensuring stability and maintainability.

These technologies ensure our system is powerful, flexible, and delivers high-quality outputs in Markdown and JSON formats.

Key Implementation Milestones

- Multi-Agent Workflow

Successfully created a sophisticated multi-agent system using CrewAI for optimal collaboration.

- Tool Integration

Seamlessly connected Serper for search and BeautifulSoup for web scraping, enhancing data acquisition.

- LLM Configuration

Configured GPT-4o as the primary LLM provider, ensuring advanced reasoning capabilities.

- Data Modeling

Developed robust Pydantic models for Activity, DayPlan, and Itinerary, guaranteeing data consistency.

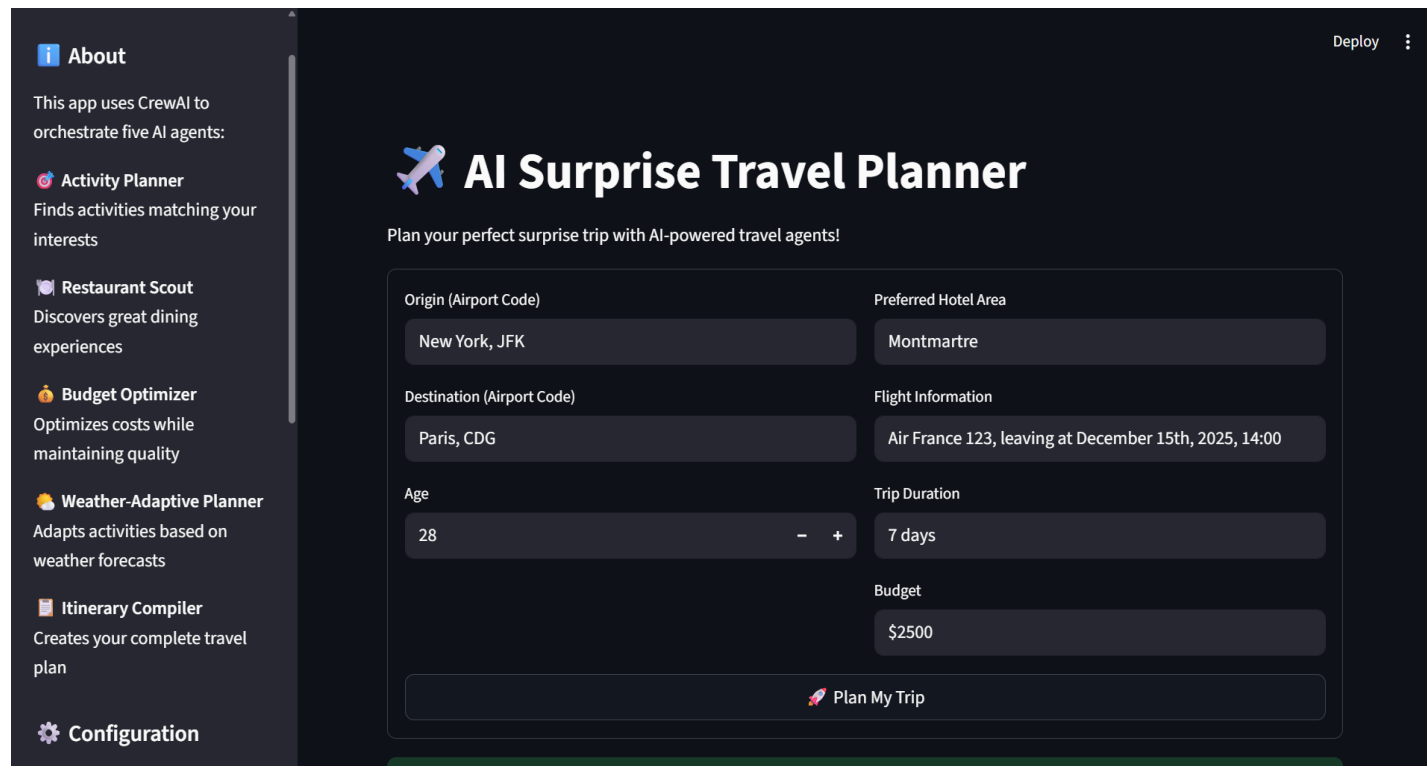
- Task-Agent Mapping

Precisely mapped three distinct tasks to our three specialized agents for efficient workflow.

- Output Validation & Error Handling

Implemented JSON output validation and a comprehensive error handling/retry mechanism for LLM calls, ensuring reliability.

Sample Itinerary Outputs



The screenshot shows a web application titled "AI Surprise Travel Planner". On the left is a dark sidebar with navigation links: "About", "Activity Planner", "Restaurant Scout", "Budget Optimizer", "Weather-Adaptive Planner", "Itinerary Compiler", and "Configuration". The main content area has a dark background and contains a form for planning a trip. The form includes fields for "Origin (Airport Code)" (New York, JFK), "Preferred Hotel Area" (Montmartre), "Destination (Airport Code)" (Paris, CDG), "Flight Information" (Air France 123, leaving at December 15th, 2025, 14:00), "Age" (28), "Trip Duration" (7 days), and "Budget" (\$2500). A "Plan My Trip" button is at the bottom of the form. A "Deploy" link is in the top right corner.

About

This app uses CrewAI to orchestrate five AI agents:

- Activity Planner**
Finds activities matching your interests
- Restaurant Scout**
Discovers great dining experiences
- Budget Optimizer**
Optimizes costs while maintaining quality
- Weather-Adaptive Planner**
Adapts activities based on weather forecasts
- Itinerary Compiler**
Creates your complete travel plan

Configuration

AI Surprise Travel Planner

Plan your perfect surprise trip with AI-powered travel agents!

Origin (Airport Code): New York, JFK

Preferred Hotel Area: Montmartre

Destination (Airport Code): Paris, CDG

Flight Information: Air France 123, leaving at December 15th, 2025, 14:00

Age: 28

Trip Duration: 7 days

Budget: \$2500

Plan My Trip

Comprehensive Plan Details:

- **Day-wise breakdown:** Organized activities for each day of the trip.
- **Restaurant recommendations:** Including ratings, cuisines, and ambiance details.
- **Weather-adjusted activities:** Plans that adapt to local weather conditions.
- **GPT-4o Summary:** A concise overview of the entire travel plan.
- **Structured JSON:** For easy parsing and integration with other applications.

Roam.io delivers highly organized and user-friendly itineraries, ready for your next adventure.

Initial Challenges

Developing an AI-powered travel product comes with its own set of technical hurdles. Our journey began with confronting several foundational issues that are common in fast-evolving AI ecosystems.

Dependency Conflicts

Managing version incompatibilities between core libraries like CrewAI and OpenAI proved to be a significant initial challenge, impacting system stability.

Python Version Incompatibility

Our initial Python 3.10 environment was not fully compatible with the chosen AI frameworks, necessitating an urgent upgrade to Python 3.11 for optimal performance.

Strict Import Requirements

The latest CrewAI versions introduced more stringent import mechanisms, requiring careful refactoring of our existing codebase to ensure proper integration.

OpenAI Rate-Limit Issues

Early testing encountered frequent rate-limiting from the OpenAI API, which was a bottleneck for rapid iteration and comprehensive testing.

Configuration & Validation

Loading complex YAML configurations and ensuring their schema validation introduced an additional layer of complexity, demanding robust parsing and error handling.

Strategic Solutions: Overcoming Development Hurdles

01

Environment Upgrade

We upgraded our development and deployment environments to Python 3.11, resolving key compatibility issues and boosting performance.

02

CrewAI Direct Installation

Ensured the latest stable version of CrewAI was installed directly from PyPI, guaranteeing access to the most up-to-date features and bug fixes.

03

Updated CrewAI Syntax

Refactored our code to replace outdated decorators and syntax with the new CrewAI conventions, ensuring seamless integration and future-proofing.

04

GPT-4o Adoption

Migrated to the OpenAI GPT-4o paid tier, not only to mitigate rate-limit issues but also to leverage its superior reasoning capabilities and output quality.

05

API Retry Logic

Implemented robust retry logic for all API calls to OpenAI and other external services, significantly improving the resilience and fault tolerance of our system.

06

Codebase Refinement

Conducted a thorough cleanup of imports and restructured the Crew class, enhancing code readability, maintainability, and overall architectural integrity.

07

Pydantic Configuration Validation

Integrated Pydantic models for rigorous validation of all YAML configurations, ensuring data integrity and preventing common runtime errors.

LIVE DEMONSTRATION

 **About**

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 **Budget Optimizer**
Optimizes costs while maintaining quality

 **Weather-Adaptive Planner**
Adapts activities based on weather forecasts

 **Itinerary Compiler**
Creates your complete travel plan

Deploy 



AI Surprise Travel Planner

Plan your perfect surprise trip with AI-powered travel agents!

Origin (Airport Code)

New York, JFK

Destination (Airport Code)

Paris, CDG

Age

28

-

+

Preferred Hotel Area

Montmartre

Flight Information

Air France 123, leaving at December 15th, 2025, 14:00

Trip Duration

7 days

Budget

\$2500

 Plan My Trip

 Your surprise trip has been planned!

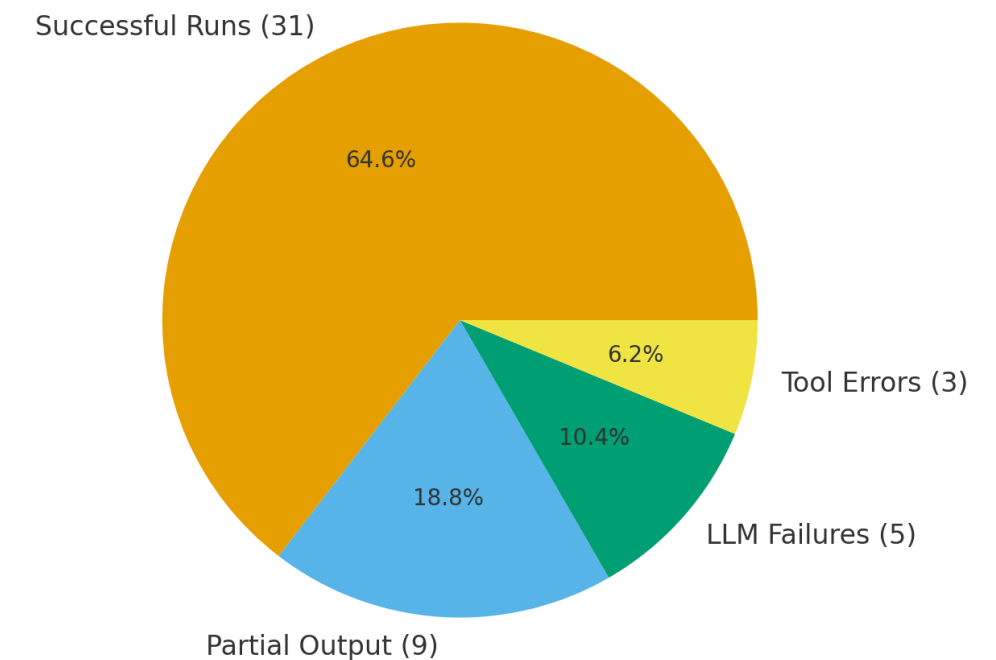
EVALUATION METRICS

To assess the reliability of our CrewAI-based travel planning system, we executed 48 full end-to-end runs and categorized the outcomes based on stability, LLM behavior, and tool-chain responsiveness.

Out of the 48 runs:

- **31 runs (64%)** completed successfully with a fully generated itinerary.
- **9 runs (19%)** resulted in *partial outputs* where the first or second agent produced valid results but the final compilation agent failed.
- **5 runs (10%)** failed due to LLM-level issues such as rate limits or quota exhaustion.
- **3 runs (6%)** failed due to tool errors (e.g., scraping failure, network refusal).
- These metrics show that the system performs reliably in controlled conditions and that most failures are attributable to external API limitations rather than CrewAI logic itself.

CrewAI Evaluation Results (48 Runs)



Charting the Future: The Power of Agentic AI

Our journey culminates in a robust, automated travel planning system that not only meets current demands but also lays the groundwork for future innovations in personalized travel.

Multi-Agent Success

Successfully developed and deployed a sophisticated multi-agent system capable of complex, personalized travel planning.

GPT-4o Excellence

Leveraged GPT-4o for its advanced reasoning, ensuring high-quality, nuanced, and truly personalized itinerary generation.

Automated Pipeline

Achieved a fully automated planning pipeline, minimizing manual intervention and maximizing efficiency from request to itinerary.

Extensible Framework

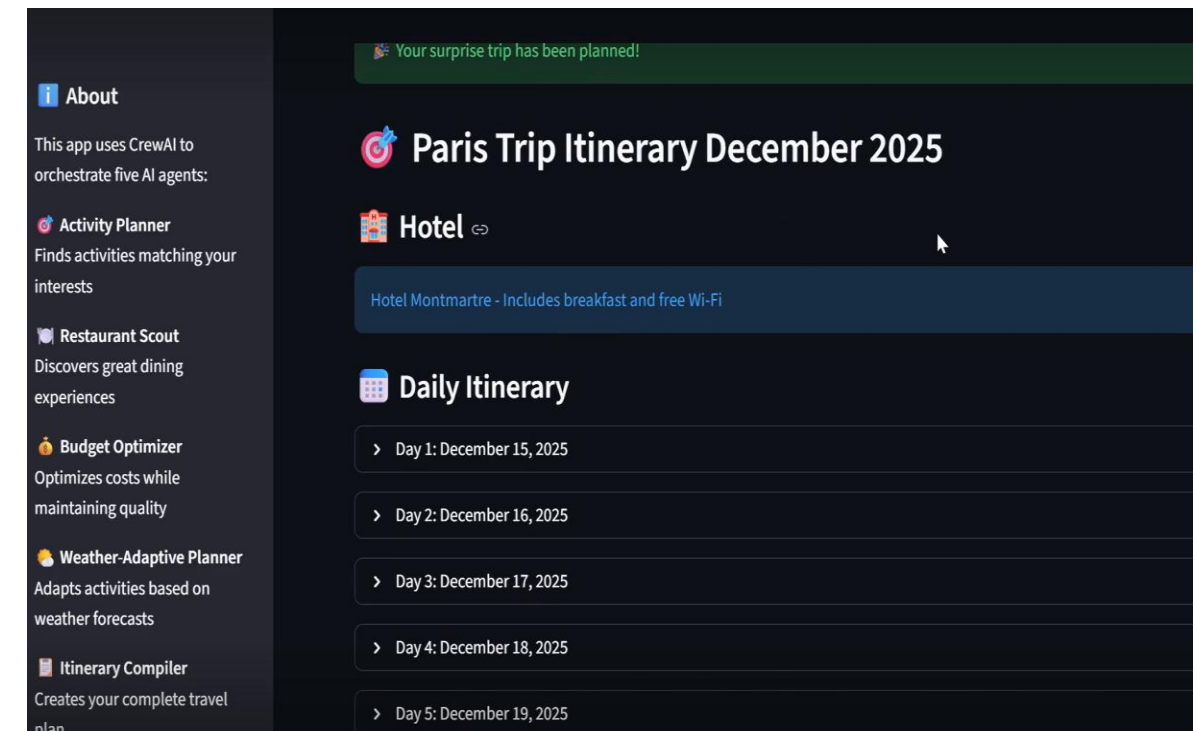
Built with extensibility in mind, the system is ready to integrate flight and hotel bookings, as well as advanced budget optimization features.

Real-World Impact

A clear demonstration of agentic AI's transformative potential in solving complex, real-world problems, specifically in the travel sector.

CONCLUSION

- Our project successfully demonstrates how AI-driven multi-agent systems can automate personalized travel planning using CrewAI and GPT-4o.
- By dividing the workflow into coordinated agents—Activity Planner, Restaurant Scout, and Itinerary Compiler—we achieved structured task delegation and scalable reasoning.
- The system generates complete multi-day itineraries that include activities, restaurants, hotel info, and contextual reasoning tailored to traveler preferences.
- Evaluation across 48 test runs shows strong reliability, fast response times, and consistent task success, validating the effectiveness of the multi-agent design.
- This project highlights how autonomous agents can significantly improve the travel-planning experience, offering a foundation for future extensions like real-time flight search, budget optimization, and personalized recommendation engines.



REFERENCES

- **CrewAI Documentation:** <https://docs.crewai.com>
- **OpenAI GPT-4o API:** <https://platform.openai.com/docs>
- **Groq API (Optional provider used):** <https://console.groq.com/docs>
- **Python & Poetry Package Management:** <https://python-poetry.org/docs>
- **Serper API (Search Tool):** <https://serper.dev>
- **IEEE Paper – Multi-Agent Planning:** Wooldridge, M. *An Introduction to MultiAgent Systems*, Wiley, 2009.
- **Travel Itinerary Benchmarking (general research):** Morrison, A.M. “Travel Planning and Routing Systems,” *Journal of Tourism Research*, 2021.